Comparison of Lumbar Disc Degeneration Grading Between Deep Learning Model SpineNet and Radiologist: A Longitudinal Study with a 14-Year Follow-Up

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Inventory Category:	External validation / generalizability
Comparability / Priority:	High | Priority: Important

Study Details & Methodology from Literature Inventory

Study Type / MRI View:	Longitudinal validation | Lumbar MRI
Dataset & Cohort:	None (Sample: None)
Disease / Target:	Disc degeneration progression
Model / AI Method:	SpineNet
Evaluation Metrics:	None
Key Finding / Relevance:	None
Thesis Context (Ch 2):	Tests stability of automated grading over time, not merely one-time test accuracy.

Abstract / Summary

Abstract Purpose To assess the agreement between lumbar disc degeneration (DD) grading by the convolutional neural network model SpineNet and radiologist's visual grading. Methods In a 14-year follow-up MRI study involving 19 male volunteers, lumbar DD was assessed by SpineNet and two radiologists using the Pfirrmann classification at baseline (age 37) and after 14 years (age 51). Pfirrmann summary scores (PSS) were calculated by summing individual disc grades. The agreement between the first radiologist and SpineNet was analyzed, with the second radiologist's grading used for inter-observer agreement. Results Significant differences were observed in the Pfirrmann grades and PSS assigned by the radiologist and SpineNet at both time points. SpineNet assigned Pfirrmann grade 1 to several discs and grade 5 to more discs compared to the radiologists. The concordance correlation coefficients (CCC) of PSS between the radiologist and SpineNet were 0.54 (95% CI: 0.28 to 0.79) at baseline and 0.54 (0.27 to 0.80) at follow-up. The average kappa (κ) values of 0.74 (0.68 to 0.81) at baseline and 0.68 (0.58 to 0.77) at follow-up. CCC of PSS between the radiologists was 0.83 (0.69 to 0.97) at baseline and 0.78 (0.61 to 0.95) at follow-up, with κ values ranging from 0.73 to 0.96. Conclusion We found fair to substantial agreement in DD grading between SpineNet and the radiologist, albeit with notable discrepancies. These findings indicate that AI-based systems like SpineNet hold promise as complementary tools in radiological evaluation, including in longitudinal studies, but emphasize the need for ongoing refinement of AI algorithms.